

SmartBin-1

Ranging · Level-shifting · Calibration · Error & Power /

Automated IoT Waste-Level Monitoring & Alert System · Prototype V1.0

REPORT DATE

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BASIS

Project report + prototype tests

±2.88 %

WORST-CASE SENSOR ERROR

3.33 V

ECHO LEVEL AT GPIO18

6.4 mA

MAX LED CURRENT

≈ 5 Hz

SENSING RATE (201 ms)

01 DESIGN INPUTS & ASSUMPTIONS

Symbol	Quantity	Value	Basis
D_{empty}	Calibrated sensor-to-bottom depth	14.4 cm	Project report / firmware
D_{full}	Distance at 100 % fill (headroom above min range)	4.0 cm	Project report / firmware
S	Usable span = $D_{\text{empty}} - D_{\text{full}}$	10.4 cm	Calculated
v	Speed of sound used by firmware ($\approx 20^\circ\text{C}$)	343 m/s = 0.0343 cm/ μs	Firmware constant
$\pm e$	HC-SR04 nominal ranging accuracy	± 0.3 cm	Sensor datasheet (nominal)
$R1 / R2$	ECHO divider ($R2 = \text{two } 1 \text{ k}\Omega \text{ in series}$)	1 k Ω / 2 k Ω , 5 % tol.	Bill of materials
R_{LED}	LED series resistor ($\times 3$)	220 Ω	Bill of materials
$V_{\text{cc}} / V_{\text{io}}$	Sensor & buzzer rail / ESP32 GPIO level	5 V (USB/VIN) / 3.3 V	ESP32 DevKit V1
$t_{\text{out}}', t_{\text{loop}}', t_{\text{poll}}$	Echo timeout / loop delay / dashboard poll	30 ms / 200 ms / 500 ms	Firmware & dashboard

02 ULTRASONIC RANGING

The HC-SR04 raises ECHO for the round-trip flight time t of a 40 kHz burst. The one-way distance and its inverse are:

$$D = t \cdot v / 2 = t \times 0.0343 / 2 \quad \Leftrightarrow \quad t = 2D / v = 2D / 0.0343$$

Condition	D (cm)	Echo time t (μs)	Working
Empty bin (0 %)	14.4	839.7	$2 \times 14.4 / 0.0343$
Half-full alert (50 %)	9.2	536.4	$2 \times 9.2 / 0.0343$
Full alert (80 %)	6.08	354.5	$2 \times 6.08 / 0.0343$
Full bin (100 %)	4.0	233.2	$2 \times 4.0 / 0.0343$
Firmware lower clamp	2.0	116.6	$2 \times 2.0 / 0.0343$

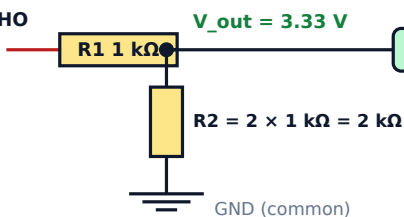
Resolution: pulseIn() counts in 1 μs steps $\rightarrow \Delta D = 0.0343 / 2 = \mathbf{0.01715 \text{ cm}}$ per tick = 0.165 % of the 10.4 cm span (negligible).

Timeout: 30 000 μs corresponds to $30\,000 \times 0.0343 / 2 = 514.5$ cm, beyond the sensor's rated range, so the timeout only catches a lost echo (firmware then reports the bin as empty).

03 ECHO LEVEL-SHIFTING (VOLTAGE DIVIDER)

HC-SR04 ECHO

5 V pulse



The HC-SR04 drives ECHO at 5 V but ESP32 GPIOs are 3.3 V devices. A 1 k Ω / 2 k Ω divider scales the pulse:

$$V_{\text{out}} = V_{\text{in}} \cdot R2 / (R1 + R2) = 5 \times 2 \text{ k} / 3 \text{ k} = 3.33 \text{ V}$$

Divider current $I = 5 / 3 \text{ k} = \mathbf{1.67 \text{ mA}}$; source impedance $R1 \parallel R2 = 667 \Omega$, so with $\approx 10 \text{ pF}$ pin capacitance (assumed) $\tau \approx 6.7 \text{ ns}$ — negligible against the 1 μs timing resolution.

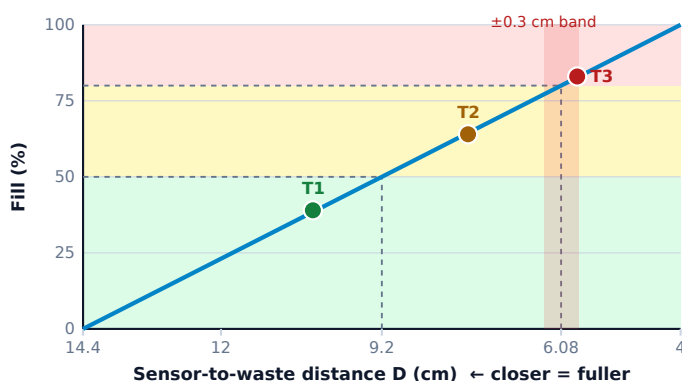
Case	R1	R2	V _{out}	Check ($V_{\text{IH}} \approx 0.75 V_{\text{DD}} = 2.48 \text{ V}$; GPIO max $\approx 3.6 \text{ V}$)
Nominal	1.00 k Ω	2.00 k Ω	3.333 V	HIGH recognised ✓ · within limit ✓
Worst low ($\pm 5 \%$)	1.05 k Ω	1.90 k Ω	3.220 V	$5 \times 1.9 / 2.95 \rightarrow$ HIGH recognised ✓
Worst high ($\pm 5 \%$)	0.95 k Ω	2.10 k Ω	3.443 V	$5 \times 2.1 / 3.05 \rightarrow$ below 3.6 V ✓

Case	R1	R2	V_out	Check ($V_{IH} \approx 0.75 V_{DD} = 2.48\text{ V}$; GPIO max $\approx 3.6\text{ V}$)
No divider (direct)	—	—	5.00 V	1.4 V above GPIO limit <i>X</i> (stress / damage risk)

04 FILL CALIBRATION & ALERT THRESHOLDS

Fill is linear in distance over the 10.4 cm span (firmware: `round()` then `constrain(0,100)`):

$$\text{Fill \%} = (D_{\text{empty}} - D) / (D_{\text{empty}} - D_{\text{full}}) \times 100 = (14.4 - D) / 10.4 \times 100 \quad | \quad D_{x\%} = 14.4 - 0.104 \cdot x$$



Fill	D (cm)	Status	LED	Buzzer
< 50 %	> 9.2	OK	Green · GPIO14	OFF
50–79 %	6.08 – 9.2	HALF FULL	Yellow · GPIO27	OFF
≥ 80 %	≤ 6.08	FULL	Red · GPIO26	ON · 2500 Hz

Thresholds: 50 % → $14.4 - 5.2 = 9.2$ cm; 80 % → $14.4 - 8.32 = 6.08$ cm. 1 % of fill = 0.104 cm of distance. Shaded strip = ±0.3 cm sensor accuracy around the 80 % alarm point.

Verification against prototype tests

Test	D shown	Calculated fill	Reported	Δ	State · indicators
T1	10.4 cm	$(14.4 - 10.4) / 10.4 = 38.5$ %	39 %	+0.5 %	OK · green LED, buzzer off
T2	7.7 cm	$(14.4 - 7.7) / 10.4 = 64.4$ %	64 %	−0.4 %	HALF FULL · yellow LED, buzzer off
T3	5.8 cm	$(14.4 - 5.8) / 10.4 = 82.7$ %	83 %	+0.3 %	FULL · red LED, buzzer on

Differences are rounding only (the dashboard shows distance to 0.1 cm while the device computes fill from the unrounded reading).

05 LED CURRENT-LIMIT DESIGN

$$I = (V_{\text{GPT0}} - V_F) / R \quad | \quad R_{\text{min}} = (3.3 - V_F) / I_{\text{max}} \quad | \quad P_R = I^2 R$$

LED	Typical V_F^*	I at 220 Ω	P in resistor	Min R for 12 mA	Check
Green (3 mm)	2.2 V	$(3.3 - 2.2) / 220 = 5.0$ mA	5.5 mW	92 Ω	✓ safe
Yellow (3 mm)	2.0 V	$(3.3 - 2.0) / 220 = 5.9$ mA	7.7 mW	108 Ω	✓ safe
Red (5 mm)	1.9 V	$(3.3 - 1.9) / 220 = 6.4$ mA	8.9 mW	117 Ω	✓ safe

*Typical forward voltages assumed, not measured. Only one LED is lit at a time, so peak LED current is 6.4 mA — about half of a conservative 12 mA per-pin budget and far below the 250 mW resistor rating. 220 Ω gives visible brightness with large margin.

06 ERROR ANALYSIS

$$\text{Error}_{\text{fill}} = \pm e / S \times 100 = \pm (0.3 / 10.4) \times 100 = \pm 2.88 \text{ \%} \quad (\text{sensor accuracy, ideal line of sight, } \approx 20 \text{ }^\circ\text{C})$$

Error source	Magnitude	Effect on fill	Status
HC-SR04 accuracy	±0.3 cm	±2.88 %	Dominant, as reported
Timer quantisation (1 μs)	0.017 cm	±0.08 %	Negligible
Integer rounding of fill	—	±0.5 %	Display only
Speed of sound at 20 °C (343.4 vs 343)	+0.12 %	≤ 0.17 % at D = 14.4	Negligible
Speed of sound at 30 °C (349.5 m/s)	+1.88 %	≈ 2.6 % at D = 14.4	Not compensated
Speed of sound at 40 °C (355.5 m/s)	+3.65 %	≈ 5.1 % at D = 14.4	Not compensated

Temperature rows use $v \approx 331.3 + 0.606 \cdot T$ (m/s). Distance error scales with D, so the effect is largest when the bin is nearly empty. Worst-case combined at 30 °C ≈ 2.88 + 2.6 ≈ 5.5 %; at 40 °C ≈ 7.9 % (linear sum). These go beyond the room-condition tests in the project report.

Threshold margin flag: Test 3 (5.8 cm) lies only 0.28 cm from the 80 % alarm point (6.08 cm) — inside the ±0.3 cm accuracy band. Near that boundary a reading can jitter between yellow and red/buzzer. A 3–5 sample median and a small hysteresis (e.g. alarm ON at 6.08 cm, OFF at 6.4 cm) would remove flicker.

07 POWER BUDGET (5 V USB SUPPLY)

Load	Typical	Peak	Note
ESP32 DevKit (Wi-Fi AP active)	120 mA	240 mA	Estimate; Wi-Fi TX bursts
HC-SR04 ultrasonic sensor	15 mA	15 mA	Typical datasheet value

Load	Typical	Peak	Note
LED (one at a time)	6.4 mA	6.4 mA	Calculated in §05
Active buzzer module (FULL state only)	30 mA	30 mA	Assumed; confirm on bench
Total	171.4 mA	291.4 mA	$P = 5\text{ V} \times I = 0.86\text{ W typ.} / 1.46\text{ W peak}$

A standard USB 2.0 port supplies 500 mA, leaving $\geq 208\text{ mA}$ ($\approx 42\%$) margin at peak. All currents here are datasheet-type estimates, not bench measurements.

08 TIMING & RESPONSIVENESS

Quantity	Working	Result
Loop period	200 ms delay + $\leq 0.84\text{ ms}$ echo + $12\text{ }\mu\text{s}$ trigger + web service $\approx 201\text{ ms}$	$\approx 5\text{ Hz}$ sensing
Local alert latency (LED / buzzer)	$\leq$ one loop period	$\leq 0.2\text{ s}$
Dashboard refresh	JavaScript polls /data every 500 ms	2 Hz
Worst-case remote display latency	500 ms poll + 201 ms sample age	$\approx 0.7\text{ s}$
Wi-Fi coverage (AP mode)	On-board antenna, line of sight (project report)	$\approx 15 - 30\text{ m}$

09 DESIGN CHECK SUMMARY

Item	Criterion	Result	Status
ECHO input voltage	$\leq 3.6\text{ V}$ and $\geq 2.48\text{ V}$	$3.22 - 3.44\text{ V}$ ($\pm 5\%$ resistors)	PASS
LED pin current	$\leq 12\text{ mA}$ per pin	6.4 mA max	PASS
Test accuracy	$ \text{calculated} - \text{reported} \leq 1\%$	$\leq 0.5\%$ (rounding)	PASS
Sensor accuracy	Report claim $\pm 2.88\%$	$\pm 2.88\%$ at $\approx 20\text{ }^{\circ}\text{C}$	PASS
Supply current	$\leq 500\text{ mA}$ (USB 2.0)	$\approx 291\text{ mA}$ peak (estimate)	PASS
80 % alarm margin	Margin $> \pm 0.3\text{ cm}$	0.28 cm at Test 3	FLAG
Temperature drift	$< 3\%$ fill	2.6% @ $30\text{ }^{\circ}\text{C}$ · 5.1% @ $40\text{ }^{\circ}\text{C}$	FLAG
Buzzer idle state	Silent below 80 %	Module is marked "low-level trigger"; silent in tests	CHECK

Recommended improvements

- Add a 3–5 sample median filter and $\sim 0.3\text{ cm}$ hysteresis at the 50 % and 80 % thresholds.
- Compensate the speed of sound with a temperature sensor: $v = 331.3 + 0.606 \cdot T$.
- Bench-measure LED forward voltage and total supply current to replace the typical values used here.
- Re-verify the buzzer's silent idle level after any firmware change (low-level-trigger module).